🎯 Gold Scalping Strategy — Consolidated Indicator Analysis & Blueprint

> **Purpose:** Merge, fix, and optimize 4 Pine Script indicators into a single profitable **Gold (XAU/USD) scalping** system. Includes Dave's Quarter Theory logic specifically tuned for gold's psychological price behavior.

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Indicator 1: Trendline Breakout

Original Issues

Severity	Issue	Details
Medium	Division by zero risk in `linreg_coefs()`	`n * sumX2 - sumX * sumX` can numerically underflow
Medium	`var` state persistence without reset	`s_val`, `r_val`, `sig` persist stale values before `lookback + 1` bars
Low	No neutral zone	`sig` holds 1 or -1 continuously —no "between lines" state
Low	Hardcoded `1e-5` tolerance	May discard legitimate micro-violations on low-volatility instruments

Fixes Applied

```
// Fix 1: Add initialization guard
if bar_index == 0
    s_val := na
    r_val := na
    sig := 0

// Fix 2: Safer linreg with zero-denominator guard
float denom = n * sumX2 - sumX * sumX
float slope = denom != 0 ? (n * sumXY - sumX * sumY) / denom : 0.0
float intercept = denom != 0 ? (sumY - slope * sumX) / n : src[len / 2]

// Fix 3: Add ATR-based confirmation for breakouts (reduces false signals)
```

```
float atr = ta.atr(14)
bool confirmed_break_up = close > r_val + atr * 0.15
bool confirmed_break_down = close < s_val - atr * 0.15
```

How to Use for Gold Scalping

- Set `lookback = 24` to `36` for M5/M15 gold charts (gold moves fast)
- Use the **non-repainting** `[1]` lag for all calculations — critical for backtest validity
- The OLS + pivot optimization creates dynamic S/R that adapts to gold's volatility clusters

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Indicator 2: BigBeluga Smart Money Concepts

Original Issues

Severity	Issue	Details
Critical	Memory leak in `barstate.islast` cleanup	Objects created every tick, deleted only on last confirmed bar
Critical	Division by zero risk	`atr = ta.atr(200) / (5/len)` — integer division makes `5/len = 0` when `len > 5`
High	Array index out of bounds in `fnOB`	`idx` from `find()` can exceed available history
High	Race condition in FVG raid detection	Same bar can trigger both `israid` and `active` states
High	Inefficient `structure()` function	`var` arrays inside function grow unbounded; $O(n^2)$ loops
Medium	Phantom state in `entered` types	`blobenter.normal` is global, not per-block —suppresses all subsequent blocks
Medium	No state reset for `var` variables	Stale values from previous chart loads may linger

Fixes Applied

```
// Fix 1: ATR calculation (CRITICAL — was causing division by zero)
float atr = ta.atr(200) * len / 5.0  // Use float literal 5.0

// Fix 2: Object lifecycle management
if barstate.islast
    for obj in bin.ln
        obj.delete()
    // ... clear ALL arrays
    bin.ln.clear()
    bin.lb.clear()
    bin.bx.clear()
    bin.lf.clear()

```

```
// Fix 3: Add bar_index == 0 reset
if bar_index == 0
    // Reset all var structures
    ms := structure.new(start = 0)
    blob.clear()
    brob.clear()

// Fix 4: Per-block entry tracking (replace global `entered` with block-local flags)
type ob
    bool  bull
    float top
    float btm
    float avg
    int   loc
    color css
    float vol
    int   dir
    int   move
    int   blPOS
    int   brPOS
    int   xlocbl
    int   xlocbr
    bool  isbb   = false
    int   bbloc
    bool  entered_normal  = false   // NEW: per-block state
    bool  entered_breaker = false   // NEW: per-block state
```

How to Use for Gold Scalping

- **Order Blocks**: Gold respects 15M/1H order blocks extremely well due to institutional flow
- **FVGs**: Use 5M FVGs for scalp entries, 1H FVGs for swing targets
- **Market Structure**: CHoCH on M5 + BOS on M15 = high-probability gold reversal
- **Disable** the `mapping()` function for scalping — too slow, use only for higher timeframes

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Indicator 3: SpacemanBTC Key Levels

Original Issues

Severity	Issue	Details
Critical	`lookahead=barmerge.lookahead_on` on ALL `request.security()` calls	Causes repainting —backtests show impossible fills
High	Weekly low line uses `weekly_open` instead of `weeklyl_open`	Copy-paste error —plots open price as low
High	Monthly high line uses `monthlyl_time` instead of `monthlyh_time`	Wrong time anchor for monthly high

Table 3 – continued

Severity	Issue	Details
High	Monday range captures current day, not actual Monday	Uses `cdailyh_open` (current day) instead of Monday' s range
Medium	Session low initialized to `close` instead of `low`	`clondonlow = close` should be `low` or a large number
Medium	`f_LevelMerge` uses exact float equality	Misses near-matching levels due to floating-point representation
Medium	`var` objects inside `barstate.islast`	Dangling references when new bar forms

Fixes Applied

```
// Fix 1: Remove lookahead_on for all same-timeframe or historical data
[daily_time, daily_open] = request.security(syminfo.tickerid, 'D', [time, open],
↪  lookahead=barmerge.lookahead_off)

// Fix 2: Fix weekly low line
var weeklyl_line = line.new(x1=weeklyl_time, x2=weeklyl_limit_right, y1=weeklyl_open, y2=weeklyl_open,
↪  ...)
// Was: y1=weekly_open, y2=weekly_open §$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/
↪  share/docminer/emoji/274c.png}}}}}}$§

// Fix 3: Fix monthly high anchor
line.set_x1(monthlyh_line, monthlyh_time)  // Was: monthlyl_time §$
↪  \vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/274c.png}}}}}}$§

// Fix 4: Fix Monday range logic
if dayofweek == dayofweek.monday and untested_monday == false
    untested_monday := true
    monday_time := daily_time
    monday_high := high[1]   // Previous bar high (confirmed)
    monday_low := low[1]     // Previous bar low (confirmed)

// Fix 5: Session range initialization
var clondonlow = 10e20  // Was: close §$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/
↪  share/docminer/emoji/274c.png}}}}}}$§

// Fix 6: Tolerance-based level merge
method includes_approx(float[] arr, float val, float tol) =>
    bool found = false
    for item in arr
        if math.abs(item - val) < tol
            found := true
            break
    found

// Fix 7: Move var declarations to global scope
```

```
var line londonh_line = na
var label londonh_label = na
// ... then update inside barstate.islast
```

How to Use for Gold Scalping

- **Daily Open**: Gold LOVES the daily open — major reversals happen within 30 min of NY open
- **Previous Day High/Low**: Most important levels for gold scalping — use 15M/30M retests
- **Session Ranges**: London session high/low + NY session open = gold's most volatile window
- **Quarter Theory Alignment**: Round numbers and .00/.50/.25/.75 are psychological magnets for gold

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Indicator 4: Victor Aimstar Multi-Filter Strategy

Original Issues

Severity	Issue	Details
Critical	Signal count logic is broken	`leadinglong_count` loop resets but `leadinglong_count2` retains stale value
Critical	QQE Mod short condition bug	`leadingshortcond := isqqeabove` should be `isqqebelow`
High	Same-timeframe `request.security` lag	EMAs and Stochastic wrapped in redundant `request.security` calls
High	`kernel_regression size` always = 1	`array.size(array.from(rqksrc))` returns 1, not series length
High	Supply/Demand deleted box references not removed from arrays	`box.delete()` doesn' t remove from array —causes stale references
Medium	Sydney DST logic incorrect	`sydDST` boundary months wrong
Medium	`detectCrossWithLookback` doesn' t verify most recent cross	Can return stale cross from deeper history
Medium	Dashboard table `var` prevents dynamic resizing	Changing confirmation indicators causes index out of bounds
Medium	Volume lower-TF arrays not null-checked	Empty arrays from `request.security_lower_tf` cause `na` cascade
Low	`CondIni` state machine race condition	Updates after signal evaluation in same bar

Fixes Applied

```
// Fix 1: Replace broken signal count with bar-index tracking
var int last_leading_long_bar = na
var int last_leading_short_bar = na
```

```
if leadinglongcond
    last_leading_long_bar := bar_index
if leadingshortcond
    last_leading_short_bar := bar_index

bool longcond_withexpiry = longCond and (bar_index - last_leading_long_bar <= signalexpiry)
bool shortcond_withexpiry = shortCond and (bar_index - last_leading_short_bar <= signalexpiry)

// Fix 2: QQE Mod short condition
else if leadingindicator == 'QQE Mod'
    leadinglongcond := isqqeabove
    leadingshortcond := isqqebelow  // Was: isqqeabove §$
    ↪ \vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
    ↪ 274c.png}}}}}}$§

// Fix 3: Remove same-timeframe request.security
// BEFORE: ema1 := request.security(syminfo.tickerid, timeframe.period, ma_function(...))
// AFTER:  ema1 := ma_function(close, len1, ma_1_type)

// Fix 4: Fix RQK kernel size
// BEFORE: size = array.size(array.from(rqksrc))
// AFTER:  size = int(h2)  // Use the input lookback window directly

// Fix 5: Clean SD zone arrays after deletion
f_sd_to_bos(box_array, bos_array, label_array, zone_type) =>
    // ... after box.delete(array.get(box_array, i))
    array.remove(box_array, i)
    array.remove(label_array, i)
    // Decrement i or break if only processing first match

// Fix 6: Fix Sydney DST
bool sydDST = (month > 4 and month < 10) ? false : true  // Simplified: DST = Oct-Mar

// Fix 7: Fix detectCrossWithLookback — verify most recent cross only
detectCrossWithLookback(fast, slow, lookback) =>
    int crossBar = na
    for i = 0 to lookback
        if ta.crossover(fast[i], slow[i]) or ta.crossunder(fast[i], slow[i])
            crossBar := i
            break  // Take MOST RECENT cross
    // ... rest of logic

// Fix 8: Dashboard table — rebuild each bar instead of var
if showdashboard and barstate.islast
    var table tab1 = na
    if na(tab1) or array.size(confirmation_counter) != tab1.rows() - 4
        tab1.delete()
        tab1 := table.new(...)
```

How to Use for Gold Scalping

- **Leading Indicator**: Use `Range Filter` or `Supertrend` for gold — they handle volatility clusters well
- **Confirmation Filters** (pick 3-4 max for speed):
  1. `EMA Filter` (200 EMA — gold trends strongly)
  2. `RSI` (avoid overbought/oversold entries)
  3. `DMI/ADX` (ensure trending, not ranging)
  4. `Quarter Theory proximity` (see below)
- **Signal Expiry**: Set to `2` for M1, `3` for M5, `5` for M15 gold scalping
- **Avoid**: `Ichimoku`, `QQE Mod`, `B-Xtrender` for scalping — too laggy for gold's speed

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Dave's Quarter Theory for Gold

Core Concept

Gold prices gravitate toward **quarter points** — prices ending in `.00`, `.25`, `.50`, `.75`. These act as psychological magnets, support/resistance, and profit targets.

Quarter Theory Rules for Gold

Quarter	Psychological Level	Behavior
`.00`	Major whole number	Strongest S/R —major reversals, take-profit clusters
`.50`	Half number	Secondary S/R —often mid-point of range
`.25`	Minor quarter	Weak S/R —often breached, use as entry refinement
`.75`	Minor quarter	Weak S/R —often breached, use as entry refinement

Gold-Specific Quarter Theory Logic

```
// ── Quarter Theory Module ─────────────────────────────────────
float currentPrice = close
float wholeNumber = math.round(currentPrice)
float decimalPart = currentPrice - math.floor(currentPrice)

// Find nearest quarters
float lowerQuarter = math.floor(currentPrice * 4) / 4
float upperQuarter = math.ceil(currentPrice * 4) / 4
float midQuarter = (lowerQuarter + upperQuarter) / 2  // Usually .50 or .00

// Distance to nearest major quarter (.00 or .50)
float distToWhole = math.abs(currentPrice - wholeNumber)
```

```
float distToHalf = math.abs(currentPrice - (wholeNumber + 0.5))
float distToMajorQuarter = math.min(distToWhole, distToHalf)

// Quarter Theory Signal Generation
bool nearWholeNumber = distToWhole < atr * 0.3
bool nearHalfNumber = distToHalf < atr * 0.3
bool nearQuarter = distToMajorQuarter < atr * 0.2

// Gold Scalping Rules:
// 1. If price is within 20% ATR of .00 → expect bounce or consolidation
// 2. If price breaks .00 with volume → expect run to next .00 or .50
// 3. .50 acts as pivot — above = bullish, below = bearish
// 4. Round numbers are liquidity pools — smart money hunts stops just beyond them

// Quarter-based S/R levels for current session
float q_support = math.floor(currentPrice) + (close < math.floor(currentPrice) + 0.5 ? 0.0 : 0.5)
float q_resistance = math.ceil(currentPrice) + (close > math.ceil(currentPrice) - 0.5 ? 0.0 : 0.5)

// Gold-specific: XAU/USD often respects $5 increments ($1950, $1955, etc.)
float gold_major_increment = math.round(currentPrice / 5) * 5
float gold_minor_increment = math.round(currentPrice / 2.5) * 2.5
```

Quarter Theory + Price Action for Gold

1. **The Bounce Play**: Price approaches `.00` from above → look for bullish PA (pin bar, engulfing) → scalp long to `.25` or `.50`
2. **The Breakout Play**: Price consolidates near `.00` → breaks with momentum → enter on retest of `.00` as support → target next `.00`
3. **The Fakeout Play**: Price spikes past `.00` by $1-2 → reverses sharply → this is stop-hunting → fade the move back to `.00`
4. **The 50-Cent Pivot**: Price above `.50` = bullish bias; below `.50` = bearish bias for that $1 range

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Merged Gold Scalping Architecture

Recommended Timeframes

- **Primary**: M5 (entry/execution)
- **Confirmation**: M15 (trend direction)
- **Context**: H1 (key levels, order blocks)
- **Execution**: M1 (precision entry on quarter levels)

Module Integration

```
┌─────────────────────────────────────────────────────────────┐
│                     GOLD SCALPING SYSTEM                     │
├─────────────────────────────────────────────────────────────┤
│  LAYER 1: MARKET STRUCTURE (BigBeluga SMC)                  │
│  ├── 1H Order Blocks (supply/demand zones)                  │
│  ├── 15M FVGs (fair value gaps for entries)                 │
```

```
│  └── 15M CHoCH/BOS (trend direction)                     │
├──────────────────────────────────────────────────────────┤
│  LAYER 2: KEY LEVELS (SpacemanBTC)                        │
│  ├── Daily Open / Previous Day H/L                        │
│  ├── London & NY Session Ranges                           │
│  └── Quarter Theory Levels (.00, .25, .50, .75)          │
├──────────────────────────────────────────────────────────┤
│  LAYER 3: TREND CONFIRMATION (Victor Aimstar — Filtered)    │
│  ├── Leading: Range Filter or Supertrend (M5)            │
│  ├── Confirm 1: 200 EMA direction (M15)                   │
│  ├── Confirm 2: RSI (avoid OB/OS entries)                 │
│  └── Confirm 3: ADX > 25 (trending market)              │
├──────────────────────────────────────────────────────────┤
│  LAYER 4: PRECISION ENTRY (Trendline Breakout + Quarter)    │
│  ├── Dynamic trendlines on M5 (non-repainting)           │
│  ├── Breakout + ATR confirmation                          │
│  └── Entry within 20% ATR of quarter level                │
├──────────────────────────────────────────────────────────┤
│  RISK MANAGEMENT                                          │
│  ├── Stop: 1.5x ATR or beyond nearest OB/FVG             │
│  ├── Target 1: Next quarter level (1:1 R/R)              │
│  ├── Target 2: Next major .00 level (2:1 R/R)            │
│  └── Max 2 trades per quarter-hour candle                 │
└──────────────────────────────────────────────────────────┘
```

Signal Logic (AND Gate)

```
bool gold_long_signal =
    // Layer 1: Structure
    price_above_nearest_demand_zone and
    m15_bullish_structure and

    // Layer 2: Levels
    (price_near_quarter_level or price_at_session_open) and
    not price_at_major_resistance and

    // Layer 3: Trend
    range_filter_bullish and
    price_above_200ema and
    rsi_not_overbought and
    adx_above_25 and

    // Layer 4: Precision
    trendline_breakout_up and
    close > resistance_line + atr * 0.15

bool gold_short_signal =
    // Layer 1: Structure
    price_below_nearest_supply_zone and
    m15_bearish_structure and
```

```
    // Layer 2: Levels
    (price_near_quarter_level or price_at_session_open) and
    not price_at_major_support and

    // Layer 3: Trend
    range_filter_bearish and
    price_below_200ema and
    rsi_not_oversold and
    adx_above_25 and

    // Layer 4: Precision
    trendline_breakout_down and
    close < support_line - atr * 0.15
```

Complete Fixed & Merged Pine Script Blueprint

```
//@version=6
strategy("Gold Scalper Pro — Merged & Fixed",
     overlay=true,
     max_lines_count=500,
     max_boxes_count=500,
     max_labels_count=500,
     default_qty_type=strategy.percent_of_equity,
     default_qty_value=2.0,
     commission_type=strategy.commission.cash_per_contract,
     commission_value=0.05,
     slippage=1)

// ═══════════════════════════════════════════════════════════════
// INPUTS
// ═══════════════════════════════════════════════════════════════
group_time = "§$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
 ↪  23f0.png}}}}}}$§ Timeframes & Sessions"
group_structure = "§$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
 ↪  1f4ca.png}}}}}}$§ Market Structure (SMC)"
group_levels = "§$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
 ↪  1f3af.png}}}}}}$§ Key Levels & Quarter Theory"
group_trend = "§$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
 ↪  1f4c8.png}}}}}}$§ Trend Confirmation"
group_risk = "§$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
 ↪  1f6e1-fe0f.png}}}}}}$§ Risk Management"

// Timeframes
string tf_primary = input.string("5", "Primary TF", group=group_time)
string tf_confirm = input.string("15", "Confirmation TF", group=group_time)
string tf_context = input.string("60", "Context TF", group=group_time)

// Sessions
bool use_london = input.bool(true, "London Session (08:00-16:30 UTC)", group=group_time)
```

```
bool use_ny = input.bool(true, "NY Session (14:30-21:00 UTC)", group=group_time)
bool use_asia = input.bool(false, "Asia Session (00:00-06:00 UTC)", group=group_time)

// SMC
bool show_ob = input.bool(true, "Show Order Blocks", group=group_structure)
int ob_lookback = input.int(10, "OB Swing Lookback", minval=2, maxval=50, group=group_structure)
int ob_history = input.int(5, "OB History to Keep", minval=1, maxval=20, group=group_structure)
bool show_fvg = input.bool(true, "Show FVGs", group=group_structure)
int fvg_history = input.int(5, "FVG History", minval=1, maxval=20, group=group_structure)

// Quarter Theory
bool use_quarter_theory = input.bool(true, "Enable Quarter Theory", group=group_levels)
float quarter_tolerance = input.float(0.2, "Quarter Proximity (x ATR)", minval=0.05, maxval=0.5,
↪  step=0.05, group=group_levels)
bool show_quarter_lines = input.bool(true, "Show Quarter Lines", group=group_levels)

// Key Levels
bool show_daily_open = input.bool(true, "Daily Open", group=group_levels)
bool show_prev_day_hl = input.bool(true, "Previous Day H/L", group=group_levels)
bool show_session_ranges = input.bool(true, "Session High/Low", group=group_levels)

// Trend Filters
string leading_indicator = input.string("Range Filter", "Leading Indicator",
     options=["Range Filter", "Supertrend", "Trendline Breakout"], group=group_trend)
int rf_period = input.int(100, "RF Period", group=group_trend)
float rf_mult = input.float(3.0, "RF Multiplier", group=group_trend)
int st_period = input.int(10, "ST ATR Period", group=group_trend)
float st_mult = input.float(3.0, "ST Multiplier", group=group_trend)
int tl_lookback = input.int(24, "Trendline Lookback", minval=5, maxval=100, group=group_trend)

// Confirmation
bool use_ema200 = input.bool(true, "200 EMA Filter", group=group_trend)
bool use_rsi = input.bool(true, "RSI Filter", group=group_trend)
int rsi_length = input.int(14, "RSI Length", group=group_trend)
int rsi_overbought = input.int(70, "RSI Overbought", group=group_trend)
int rsi_oversold = input.int(30, "RSI Oversold", group=group_trend)
bool use_adx = input.bool(true, "ADX Filter", group=group_trend)
int adx_length = input.int(14, "ADX Length", group=group_trend)
int adx_threshold = input.int(25, "ADX Threshold", group=group_trend)

// Risk
float risk_per_trade = input.float(1.0, "Risk % Per Trade", minval=0.1, maxval=5.0, step=0.1,
↪  group=group_risk)
float rr_target1 = input.float(1.0, "R:R Target 1", minval=0.5, maxval=3.0, step=0.1, group=group_risk)
float rr_target2 = input.float(2.0, "R:R Target 2", minval=0.5, maxval=5.0, step=0.1, group=group_risk)
int max_trades_per_hour = input.int(2, "Max Trades/Hour", minval=1, maxval=10, group=group_risk)

// ═══════════════════════════════════════════════════════════════
// UTILITIES
// ═══════════════════════════════════════════════════════════════
float atr = ta.atr(14)
```

```
float atr_5 = ta.atr(5)

// Quarter Theory Functions
float get_nearest_quarter(float price) =>
    math.round(price * 4) / 4

float get_major_quarter(float price) =>
    float whole = math.round(price)
    float half = whole + (price > whole ? 0.5 : -0.5)
    math.abs(price - whole) < math.abs(price - half) ? whole : half

float dist_to_quarter(float price) =>
    float nearest = get_nearest_quarter(price)
    math.abs(price - nearest)

bool is_near_quarter(float price, float tolerance_atr) =>
    dist_to_quarter(price) < atr * tolerance_atr

// Safe Security Call (no lookahead)
[syminfo.tickerid, "D", [time, open, high[1], low[1]], barmerge.lookahead_off]

// ═══════════════════════════════════════════════════════════════
// LAYER 1: MARKET STRUCTURE (SMC) — Simplified for Speed
// ═══════════════════════════════════════════════════════════════

// Swing Points
float swing_high = ta.pivothigh(high, ob_lookback, ob_lookback)
float swing_low = ta.pivotlow(low, ob_lookback, ob_lookback)

// Order Blocks (Simplified — use last 2 swings)
var float last_swing_high = na
var float last_swing_low = na
var int last_swing_high_bar = na
var int last_swing_low_bar = na

if not na(swing_high)
    last_swing_high := swing_high
    last_swing_high_bar := bar_index[ob_lookback]
if not na(swing_low)
    last_swing_low := swing_low
    last_swing_low_bar := bar_index[ob_lookback]

// Demand Zone: Last bullish OB (below price)
float demand_top = last_swing_low
float demand_btm = last_swing_low - atr * 0.5

// Supply Zone: Last bearish OB (above price)
float supply_top = last_swing_high + atr * 0.5
float supply_btm = last_swing_high

// FVG Detection (3-candle imbalance)
```

```
bool fvg_up = low[1] > high[3]
bool fvg_down = high[1] < low[3]

float fvg_up_top = low[1]
float fvg_up_btm = high[3]
float fvg_down_top = low[3]
float fvg_down_btm = high[1]

// Structure: Higher High / Lower Low
bool hh = not na(swing_high) and swing_high > nz(last_swing_high[1], swing_high)
bool ll = not na(swing_low) and swing_low < nz(last_swing_low[1], swing_low)
bool hl = not na(swing_low) and swing_low > nz(last_swing_low[1], swing_low)
bool lh = not na(swing_high) and swing_high < nz(last_swing_high[1], swing_high)

bool bullish_structure = hh or hl
bool bearish_structure = ll or lh

// ═══════════════════════════════════════════════════════════════
// LAYER 2: KEY LEVELS & QUARTER THEORY
// ═══════════════════════════════════════════════════════════════

// Daily Levels (no lookahead)
[daily_time, daily_open_price, prev_day_high, prev_day_low] = request.security(syminfo.tickerid, "D",
     [time, open, high[1], low[1]], barmerge.lookahead_off)

// Session Detection
bool london_session = use_london and time(timeframe.period, "0800-1630")
bool ny_session = use_ny and time(timeframe.period, "1430-2100")
bool asia_session = use_asia and time(timeframe.period, "0000-0600")
bool active_session = london_session or ny_session or asia_session

// Session High/Low tracking
var float session_high = high
var float session_low = low

if london_session and not london_session[1]
    session_high := high
    session_low := low
else if london_session
    session_high := math.max(session_high, high)
    session_low := math.min(session_low, low)

// Quarter Theory Levels
float q_current = get_nearest_quarter(close)
float q_major = get_major_quarter(close)
float q_next_up = q_current + 0.25
float q_next_down = q_current - 0.25
float q_dist = dist_to_quarter(close)
bool near_quarter = is_near_quarter(close, quarter_tolerance)

// ═══════════════════════════════════════════════════════════════
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// LAYER 3: TREND CONFIRMATION
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// 200 EMA
float ema200 = ta.ema(close, 200)
bool above_ema200 = close > ema200
bool below_ema200 = close < ema200

// RSI
float rsi = ta.rsi(close, rsi_length)
bool rsi_ok_long = rsi < rsi_overbought
bool rsi_ok_short = rsi > rsi_oversold

// ADX
[float diplus, float diminus, float adx_value] = ta.dmi(adx_length, adx_length)
bool adx_ok = adx_value > adx_threshold

// Range Filter
float rf_smoothrng = ta.ema(ta.ema(math.abs(close - close[1]), rf_period), rf_period * 2 - 1) * rf_mult
var float rf_filt = close
rf_filt := close > nz(rf_filt[1]) ? close - rf_smoothrng < nz(rf_filt[1]) ? nz(rf_filt[1]) : close -
↪  rf_smoothrng :
            close + rf_smoothrng > nz(rf_filt[1]) ? nz(rf_filt[1]) : close + rf_smoothrng

float rf_upward = rf_filt > rf_filt[1] ? nz(rf_upward[1]) + 1 : rf_filt < rf_filt[1] ? 0 :
↪  nz(rf_upward[1])
float rf_downward = rf_filt < rf_filt[1] ? nz(rf_downward[1]) + 1 : rf_filt > rf_filt[1] ? 0 :
↪  nz(rf_downward[1])
bool rf_bullish = close > rf_filt and rf_upward > 0
bool rf_bearish = close < rf_filt and rf_downward > 0

// Supertrend
float st_atr = ta.atr(st_period)
float st_up = hl2 - st_mult * st_atr
float st_dn = hl2 + st_mult * st_atr
var float st_upper = st_up
var float st_lower = st_dn
st_upper := close[1] > st_upper[1] ? math.max(st_up, st_upper[1]) : st_up
st_lower := close[1] < st_lower[1] ? math.min(st_dn, st_lower[1]) : st_dn
var int st_trend = 1
st_trend := st_trend == -1 and close > st_lower[1] ? 1 : st_trend == 1 and close < st_upper[1] ? -1 :
↪  st_trend
bool st_bullish = st_trend == 1
bool st_bearish = st_trend == -1

// Trendline Breakout (Non-Repainting)
[slope, intercept] =>
    float sumX = 0.0, sumY = 0.0, sumXY = 0.0, sumX2 = 0.0
    for j = 0 to tl_lookback - 1
        float x = j
        float y = close[1 + tl_lookback - 1 - j]
```

```
        sumX += x
        sumY += y
        sumXY += x * y
        sumX2 += x * x
    float n = tl_lookback
    float denom = n * sumX2 - sumX * sumX
    float s = denom != 0 ? (n * sumXY - sumX * sumY) / denom : 0.0
    float ic = denom != 0 ? (sumY - s * sumX) / n : close[1]
    [s, ic]

[tl_slope, tl_intercept] = slope_intercept(close[1], tl_lookback)
float tl_value = tl_slope * tl_lookback + tl_intercept
bool tl_break_up = close > tl_value + atr_5 * 0.15
bool tl_break_down = close < tl_value - atr_5 * 0.15

// Leading Indicator Selection
bool lead_long = false
bool lead_short = false
if leading_indicator == "Range Filter"
    lead_long := rf_bullish
    lead_short := rf_bearish
else if leading_indicator == "Supertrend"
    lead_long := st_bullish
    lead_short := st_bearish
else
    lead_long := tl_break_up
    lead_short := tl_break_down

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// LAYER 4: SIGNAL GENERATION
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// Trade counter (per hour)
var int trades_this_hour = 0
var int current_hour = hour
if current_hour != hour
    trades_this_hour := 0
    current_hour := hour

// Signal Conditions
bool long_structure = close > demand_top or bullish_structure
bool short_structure = close < supply_btm or bearish_structure

bool long_level = near_quarter or (show_daily_open and math.abs(close - daily_open_price) < atr * 0.3)
bool short_level = near_quarter or (show_daily_open and math.abs(close - daily_open_price) < atr * 0.3)

bool long_confirm = (not use_ema200 or above_ema200) and (not use_rsi or rsi_ok_long) and (not use_adx
↪ or adx_ok)
bool short_confirm = (not use_ema200 or below_ema200) and (not use_rsi or rsi_ok_short) and (not use_adx
↪ or adx_ok)
```

```
bool long_signal = lead_long and long_structure and long_level and long_confirm and active_session and trades_this_hour < max_trades_per_hour
bool short_signal = lead_short and short_structure and short_level and short_confirm and active_session and trades_this_hour < max_trades_per_hour

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// EXECUTION
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// Stop Loss & Take Profit
float long_sl = math.min(low, demand_btm) - atr * 0.3
float long_tp1 = close + (close - long_sl) * rr_target1
float long_tp2 = close + (close - long_sl) * rr_target2

float short_sl = math.max(high, supply_top) + atr * 0.3
float short_tp1 = close - (short_sl - close) * rr_target1
float short_tp2 = close - (short_sl - close) * rr_target2

// Quarter-adjusted targets (magnet to nearest major quarter)
long_tp1 := get_major_quarter(long_tp1)
long_tp2 := get_nearest_quarter(long_tp2)
short_tp1 := get_major_quarter(short_tp1)
short_tp2 := get_nearest_quarter(short_tp2)

// Execute
if long_signal and strategy.position_size <= 0
    strategy.entry("Long", strategy.long, comment="GOLD LONG")
    strategy.exit("Long TP1", "Long", limit=long_tp1, stop=long_sl, qty_percent=50)
    strategy.exit("Long TP2", "Long", limit=long_tp2, stop=long_sl)
    trades_this_hour += 1

if short_signal and strategy.position_size >= 0
    strategy.entry("Short", strategy.short, comment="GOLD SHORT")
    strategy.exit("Short TP1", "Short", limit=short_tp1, stop=short_sl, qty_percent=50)
    strategy.exit("Short TP2", "Short", limit=short_tp2, stop=short_sl)
    trades_this_hour += 1

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// VISUALS
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// Quarter Lines
if show_quarter_lines and use_quarter_theory
    for i = -5 to 5
        float q_line = get_nearest_quarter(close) + i * 0.25
        bool is_major = math.round(q_line * 2) == q_line * 2  // .00 or .50
        line.new(bar_index - 50, q_line, bar_index, q_line,
             color=is_major ? color.new(color.white, 70) : color.new(color.gray, 85),
             style=line.style_dotted, extend=extend.right)

// Session Background
```

```
bgcolor(london_session ? color.new(color.blue, 90) : ny_session ? color.new(color.green, 90) : na)

// Structure Labels
plotshape(bullish_structure, "Bullish Structure", style=shape.triangleup, location=location.belowbar,
 ↪  color=color.green, size=size.tiny)
plotshape(bearish_structure, "Bearish Structure", style=shape.triangledown, location=location.abovebar,
 ↪  color=color.red, size=size.tiny)

// EMA
plot(ema200, "200 EMA", color=color.orange, linewidth=2)

// Signals
plotshape(long_signal, "Buy", style=shape.labelup, location=location.belowbar, color=color.green,
 ↪  text="BUY", textcolor=color.white, size=size.small)
plotshape(short_signal, "Sell", style=shape.labeldown, location=location.abovebar, color=color.red,
 ↪  text="SELL", textcolor=color.white, size=size.small)
```

Gold-Specific Optimization Notes

Why This Works for Gold

1. **Quarter Theory**: Gold is heavily traded by algorithms and institutions that place orders at round numbers. `.00` and `.50` are genuine liquidity pools.
2. **London/NY Session Overlap**: 14:30-16:30 UTC is gold's highest volatility window. The system only trades during active sessions.
3. **ATR-Based Stops**: Gold's ATR on M5 ranges from $2-$8 depending on volatility. Fixed-pip stops fail; ATR adapts.
4. **Non-Repainting Trendlines**: Critical for gold because it makes sharp moves — repainting indicators would show perfect entries that never existed in real-time.
5. **SMC Order Blocks**: Gold is driven by institutional flow. Order blocks from H1/15M are highly respected.
6. **Range Filter Leading**: Gold trends in bursts then consolidates. The Range Filter adapts to volatility better than fixed-period MAs.

Suggested Parameter Tweaks by Gold Volatility Regime

Regime	ATR (M5)	RF Mult	ST Mult	Lookback	Notes
Low Vol	<$2.00	2.5	2.5	36	Tighter stops, more signals
Normal	$2-4	3.0	3.0	24	Default settings
High Vol	$4-7	4.0	4.0	16	Wider stops, fewer signals
Extreme	>$7	5.0	5.0	12	Only strongest setups

Backtesting Checklist

- Test on XAU/USD M5 data with at least 6 months of history
- Verify no repainting: signals must appear on the candle they trigger
- Check drawdown during NFP/FOMC (consider disabling 30 min before/after)
- Validate quarter theory: count how many trades hit .00/.50 vs miss
- Ensure max 2 trades/hour prevents overtrading during choppy conditions

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Generated from analysis of 4 Pine Script indicators with applied fixes and Dave's Quarter Theory integration.